Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/04/2015
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	BRIC: Plant Mechanical Signaling During Spaceflight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	07/02/2013	**End Date:**	12/31/2016
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 12/31/2016 per NSSC information (Ed., 12/14/15) NOTE: End date is now 12/31/2015 per NSSC information (Ed., 1/14/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Swanson, Sarah (University of Wisconsin-Madison)		
Grant/Contract No.:	NNX13AM50G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants are exquisitely sensitive to mechanical stimulation which triggers changes ranging from widespread alteration in patterns of gene expression to profound shifts in development. Indeed, so pervasive is the mechanical stimulation from internal and external forces, such as the weight of tissues and organs, to a biology that has evolved under a constant 1xG, that the resultant signaling is thought to provide a required regulatory background to control a wide range of plant developmental and physiological processes. Despite this acknowledged role in plant morphogenesis, the overall contribution of such mechanical cues to plant function remains poorly defined, largely due to the difficulties in removing the pervasive 1xG-driven mechanical signal when conducting experiments on Earth. The microgravity environment of the International Space Station (ISS) therefore provides a unique tool to both directly address this hypothesis of mechanical load-driven morphogenesis, as well as to ask what features of plant growth in space reflect disruption of this mechano-signaling regulatory network. We have therefore grown seedlings of Arabidopsis thaliana on board the ISS using the Biological Research in Canisters-Petri Dish Fixation Units (BRIC-PDFU) hardware. We are comparing their patterns of growth and gene expression to ground-based controls to ask how the well-defined "fingerprints" of mechanical response are altered in space. In addition, we are characterizing plants grown in space whose mechanical signaling has been artificially activated or inhibited by mutations in a gene (TCH2) which has an established role in the mechanical signaling system of plants. Results from such analyses will address the fundamental question of the role of mechanical loading on plant growth and development on Earth as well as probing how manipulating this system through targeting specific mechanoresponse genes might be used to tailor plant growth to thrive in the unique environment of spaceflight. This experiment has also been supplemented with a parallel geneLAB experiment to monitor the growth, development, and gene expression patterns of a series of natural variants of the plant Arabidopsis thaliana in space. These variants (ecotypes) have been chosen to span the range of likely ecotypes to be used in future spaceflight experimentation and so will provide key baseline comparisons to help understand the impact of ecotypic makeup on responses to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is addressing how mechanical sensing operates in plants using Arabidopsis thaliana (Mouse Ear Cress). The work targets the Ca2+-dependent signaling machinery and a specific element of this machinery, a gene named touch-2, also known as calmodulin-like 24 (CML24). The patterns of growth and gene expression of wild type and mutants in CML24 grown on the International Space Station (ISS) are being compared to paired plants grown on Earth. On Earth, mechanical signaling systems are known to affect plant vigor, alter their growth habit (favoring plants that are smaller and more robust), and also impact on plant defense responses to pathogens. Therefore, this work aimed at analysis of how Ca2+ signaling is involved in the altered mechanical responses induced during spaceflight should also reveal the role for this system in the more general mechanically-linked responses of plants on Earth. Importantly, this research will allow analysis of the effects of the plant's own weight on its growth and development, research only possible in the microgravity environment of spaceflight. These results should provide molecular targets for potential manipulation to tailor plant growth to varied mechanical environments, especially the relatively mechanically silent environments of spaceflight and controlled enclosed environments on Earth, such as greenhouses. This experiment has also been supplemented with a parallel geneLAB experiment to monitor the growth, development, and gene expression patterns of a series of natural variants of the plant Arabidopsis thaliana in space. The responses of these variants (ecotypes) to spaceflight will not only help further define how the altered environment of space can differentially affect even closely related varieties of plants, but also help provide important background data for the design of future spaceflight experimentation using different ecotypes of this widely adopted model plant.
Task Progress:	Flight: Preflight testing (Science and Payload Verification Tests) was successfully completed in 2014 and the payload of 4 BRIC canisters containing a total of 20 60 mm Petri dishes of seeds launched on September 21, 2014 on board SpaceX-CRS4. All on orbit operations were successfully completed and the frozen samples were returned via SpaceX's Dragon capsule, splashing down on October 25, 2014. Samples were transported to Kennedy Space Center where they were de-integrated on November 4, 2014. Analysis at this point indicated ~ 100% seed germination in all BRIC19 and geneLAB samples. The samples were shipped to The University of Wisconsin-Madison for subsequent analysis. Between November 2014 and February 2015, samples were thawed, growth parameters measured, and RNA extracted for subsequent analysis of gene expression. The RNA samples were then submitted to the University of Wisconsin-Madison Biotech Center where they underwent RNA sequencing for RNAseq analysis (measurement of the level of gene expression of all genes in each sample plant's genome). RNAseq data is currently being analyzed to detect patterns of gene expression changes in both the BRIC19 and geneLAB samples associated with spaceflight. These changes are being verified in duplicate samples by the independent technique of quantitative PCR. Parallel ground-based analysis: In parallel to the flight experiments, the ecotypes from geneLAB and the cml24 touch mutants have been monitored for their responses to touch and gravitropic stimulation on Earth. Plants expressing the GFP-based 'Yellow Cameleon Nano-65' Ca2+ sensor were generated and Ca2+ changes in response to touch and gravitropic stimulation (90° rotation) monitored by confocal ratio imaging. cml24 mutants show altered touch-induced growth and effects on Ca2+ signals. In addition, alterations in gravitropic growth between wild type, the cml24 mutants, and geneLAB lines have been quantified and are being correlated with the gene expression changes seen during spaceflight.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Swanson SJ, Barker R, Ye Y, Gilroy S. "Evaluating mechano-transduction and touch responses in plant roots." Meth Mol Biol. 2015;1309:143-50. (In Plant Gravitropism: Methods and Protocols. E.B. Blancaflor, editor. New York: Springer, 2015.) http://dx.doi.org/10.1007/978-1-4939-2697-8_12 ; PubMed PMID: 25981774 , May-2015